NCERT • GRADE 7 • MATHS PART 2

Another Peek Beyond the Point

C04 • Ganita Prakash Part II

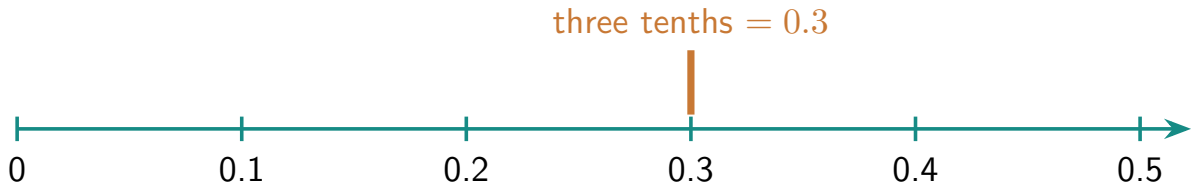
Decimal multiplication • Division

Four NCERT exercise sets with solutions

Textbook solutions first • Visual understanding • Revision

Original student notes based on NCERT. All illustrations are editable LaTeX drawings.

Study order: Four NCERT *Figure it Out* sets with worked answers first (textbook pp. 73, 83, 86–87 and 93–95); chapter explanation and revision afterward. Every numbered exercise and subpart is included. Decimal calculations are shown with their units where relevant.



1. Textbook answers: Figure it Out, page 73

Q1. Tenths and hundredths products

(a) 6×4 tenths $= 24$ tenths $= \boxed{2.4}$. **(b)** $7 \times 0.3 = 21$ tenths $= \boxed{2.1}$. **(c)** 9×5 hundredths $= 45$ hundredths $= \boxed{0.45}$.

Q2. Decimal products

(a) $27.34 \times 6 = \boxed{164.04}$ (multiply $2734 \times 6 = 16404$, then place two decimal digits).
(b) $4.23 \times 3.7 = \boxed{15.651}$ ($423 \times 37 = 15651$; total 3 decimal places).
(c) $0.432 \times 0.23 = \boxed{0.09936}$ ($432 \times 23 = 9936$; total 5 decimal places).

Q3. Cloth for three shirts

One shirt uses 1.65 m, so $3 \times 1.65 = \boxed{4.95\text{ m}}$.

Q4. Notebooks and erasers

Notebooks: $4 \times 15.50 = 62.00$ rupees. Erasers: $3 \times 2.75 = 8.25$ rupees. Total $= \boxed{70.25}$ rupees.

Q5. Height of 36 coins

Thickness $= 36 \times 1.45 = 52.2$ mm $= \boxed{5.22\text{ cm}}$. Remember 10 mm = 1 cm.

Q6. Cost of oranges

Price $= 56.50 \times 2.250 = \boxed{127.125}$ rupees (exact arithmetic).
$56.50 = 56.5$ and $2.250 = 2.25$ because trailing zeroes do not change a decimal's value; therefore both products are the same. If paying to the nearest paise, round to **Rs 127.13**.

Q7. Notebook profit

Profit per notebook $= 30 - 23.6 = 6.4$ rupees. For 50: $50 \times 6.4 = \boxed{320}$ rupees.

Q8. Use $18 \times 12 = 216$

(a) $18 \times 1.2 = 21.6$; **(b)** $18 \times 0.12 = 2.16$; **(c)** $1.8 \times 1.2 = 2.16$;
(d) $0.18 \times 0.12 = 0.0216$; **(e)** $0.018 \times 0.012 = 0.000216$; **(f)** $1.8 \times 12 = 21.6$.
Products less than 1: (d) and (e). Shift the decimal by the *total* number of decimal places in both factors.

Q9. Which products are less than 1?

(a) $7 \times 0.6 = 4.2$ (no); **(b)** $0.7 \times 0.6 = 0.42$ (yes);
(c) $0.7 \times 6 = 4.2$ (no); **(d)** $0.07 \times 0.06 = 0.0042$ (yes).
Reason without calculating: Multiplying two positive numbers each smaller than 1 produces a product smaller than 1.

Q10. Multiplication table

Number	×10	×100	×1000
5.7	57	570	5700
23.02	230.2	2302	23020
0.92	9.2	92	920
0.306	3.06	30.6	306
24.67	246.7	2467	24670

Rule: each factor of 10 shifts every digit one place left relative to the decimal point.

2. Textbook answers: Figure it Out, page 83

Q1. Divide and verify by multiplication

(a) $18/5 = \boxed{3.6}$ because $3.6 \times 5 = 18$.
(b) $415/4 = \boxed{103.75}$ because $103.75 \times 4 = 415$.
(c) $1217/2 = \boxed{608.5}$ because $608.5 \times 2 = 1217$.
(d) $4827/8 = \boxed{603.375}$ because $603.375 \times 8 = 4827$.
Multiply numerator and denominator by $2, 5, 25$ or 125 when useful to make a denominator $10, 100$ or 1000.

Q2. Choose the correct quotient

(a) $1526/4 = \boxed{381.5}$: option (iii). **(b)** $3567/8 = \boxed{445.875}$: option (iii).

Q3. The same divisor, shifting decimals

(a) $132 \div 4 = 33$; **(b)** $13.2 \div 4 = 3.3$;
(c) $1.32 \div 4 = 0.33$; **(d)** $0.132 \div 4 = 0.033$.

Q4. More place-value divisions

(a) $126/8 = 15.75$; **(b)** $12.6/8 = 1.575$; **(c)** $1.26/8 = 0.1575$;
(d) $0.126/8 = 0.01575$; **(e)** $0.0126/8 = 0.001575$.

3. Textbook answers: Figure it Out, pages 86–87

Q1. Convert fractions to decimals

(a) $\frac{2}{5} = \boxed{0.4}$; **(b)** $\frac{13}{4} = \boxed{3.25}$; **(c)** $\frac{4}{50} = \boxed{0.08}$; **(d)** $\frac{5}{8} = \boxed{0.625}$.

Q2. Decimal divisors

(a) $24.86 \div 1.2 = \frac{1243}{60} = \boxed{20.71666\ldots}$;
(b) $5.728 \div 1.52 = \frac{358}{95} = \boxed{3.7684210526\ldots}$.
Method: multiply dividend and divisor by the same power of ten to make the divisor a whole

number. Do not round unless asked.

Q3. Use $156 \times 12 = 1872$

(a) $15.6 \times 1.2 = 18.72$; **(b)** $187.2 \div 1.2 = 156$;
(c) $18.72 \div 15.6 = 1.2$; **(d)** $0.156 \times 0.12 = 0.01872$.

Q4. Missing divisors and multipliers

(a) $25 \div \boxed{1000} = 0.025$; **(b)** $25 \div \boxed{0.1} = 250$;
(c) $25 \div \boxed{10} = 2.5$; **(d)** $25 \div 10 = 25 \times \boxed{0.1}$;
(e) $25 \div 0.10 = 25 \times \boxed{10}$; **(f)** $25 \div 0.01 = 25 \times \boxed{100}$.

Q5. Effect of changing decimal divisor

(a) $2.46 \div 1.5 = \boxed{1.64}$; **(b)** $2.46 \div 0.15 = \boxed{16.4}$;
(c) $2.46 \div 0.015 = \boxed{164}$. Yes, $24.6 \div 1.5 = 16.4$, same as (b), because the dividend and divisor have both been multiplied by 10.

Q6–Q8. Equal sharing and measurement

Q6. $4 \div 5 = \boxed{0.8\,\text{m}}$ per piece.
Q7. $208.8 \div 12 = \boxed{17.4\,\text{cm}}$ per polygon side.
Q8. $3 \div 8 = 0.375\,\text{L} = \boxed{375\,\text{mL}}$ per friend.

Q9–Q10. Fuel and flour

Q9. $234.45 \div 12.6 = \boxed{18.607142857\ldots\ \text{km/L}}$ (approximately $18.61\,\text{km/L}$).
Q10. $13.5 \div 15 = \boxed{0.9\,\text{kg}}$ per student.

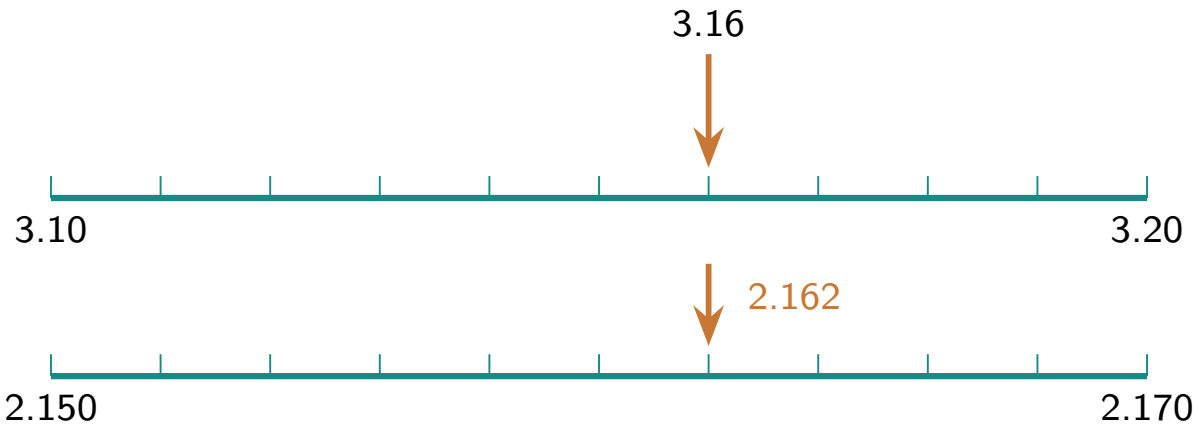
4. Textbook answers: Figure it Out, pages 93–95

Q1. Compare prices fairly

Chikki per gram $= 70.5/210 \approx 0.3357$ rupee; chips per gram $= 33.25/110 \approx 0.3023$ rupee. Hence **potato chips are cheaper by mass**. Compare unit prices, not the packet prices.

Q2. Arrow marks on the number lines

The first line runs from 3.1 to 3.2, divided into hundredths. Sixth tick after 3.1 is $\boxed{3.16}$.
The second line runs from 2.15 to 2.17, divided into ten equal parts of 0.002. Sixth tick after 2.15 is $\boxed{2.162}$.



Q3. Banana profit

Cost $= 3 \times 30 = 90$ rupees; revenue $= 35 \times 5 = 175$ rupees. Profit $= 175 - 90 = \boxed{85}$ rupees.

Q4. Bookshelf capacity

Maximum whole books $= 160 \div 2.5 = \boxed{64}$. The teacher cannot fit all 80. With 64 books the shelf is full, so **0 cm space remains**.

Q5. Convert units

(a) 5.5 km $= \boxed{5500\text{ m}}$; **(b)** 35 cm $= \boxed{0.35\text{ m}}$;
(c) 14.5 cm $= \boxed{145\text{ mm}}$; **(d)** 68 g $= \boxed{0.068\text{ kg}}$;
(e) 9.02 m $= \boxed{9020\text{ mm}}$; **(f)** 125.5 mL $= \boxed{0.1255\text{ L}}$.

Q6. Sridharacharya's divisions

First: $6\frac{1}{4} \div 2\frac{1}{2} = 6.25 \div 2.5 = \boxed{2.5}$.
Second: $60\frac{1}{4} \div 3\frac{1}{2} = 60.25 \div 3.5 = \boxed{17.214285714\ldots}$.

Q7. Two choices for each product

(a) $2.4 = \boxed{1.2 \times 2} = \boxed{0.6 \times 4}$.
(b) $14.5 = \boxed{2.9 \times 5} = \boxed{1.45 \times 10}$.

Q8. Use $756 \div 36 = 21$

(a) $75.6 \div 3.6 = 21$; **(b)** $7.56 \div 0.36 = 21$;
(c) $756 \div 0.36 = 2100$; **(d)** $75.6 \div 360 = 0.21$;
(e) $7560 \div 3.6 = 2100$; **(f)** $7.56 \div 0.36 = 21$ (same as b).

Q9. Complete the table $a \div b$

$b \backslash a$	1517	151.7	15.17	1.517	15170
37	41	4.1	0.41	0.041	410
3.7	410	41	4.1	0.41	4100
0.37	4100	410	41	4.1	41000
0.037	41000	4100	410	41	410000
370	4.1	0.41	0.041	0.0041	41

Check: Every cell is the top number divided by the left number. In particular, $15.17 \div 0.37 = 41$.

Q10. Put digits $0, 2, 4, 5, 8$ into $\square\square.\square \times \square.\square$

Use each digit once for every case.
(a) Largest: $\boxed{82.0 \times 5.4 = 442.8}$.
(b) Smallest: $\boxed{45.8 \times 0.2 = 9.16}$.
(c) Greater than 150: $24.5 \times 8.0 = 196$ (one example).
(d) Nearest to 100: $20.5 \times 4.8 = 98.4$ (difference 1.6).
(e) Nearest to 5: $45.8 \times 0.2 = 9.16$ (difference 4.16; it is also the minimum possible positive product for this layout).

Q11. Increasing order without heavy multiplication

Answer: $(f) < (c) < (a) < (e) < (d) < (b)$.
(f) 7.9682; (c) $245.05 \div 7.9682 \approx 30.75$;
(a) $245.05 \times 0.942368 \approx 230.93$; (e) 245.05;
(d) $245.05 \div 0.942368 \approx 260.04$; (b) $245.05 \times 7.9682 \approx 1952.61$.
Reason: multiplying by a positive value below 1 reduces the number; dividing by it increases the number.

5. Learn the ideas behind the answers

Decimal place value

$27.53 = 2 \times 10 + 7 \times 1 + 5 \times 0.1 + 3 \times 0.01$.
Ten tenths make one whole, and ten hundredths make one tenth. Therefore $0.50 = 0.5$—trailing zeros after the last nonzero decimal digit do not change value.

Multiplication of decimals

1. Multiply as if the numbers were whole numbers.
2. Add the decimal-place counts in both factors.
3. Place the point so the product has that many decimal places, inserting leading zeros if necessary.

Example: 0.24×0.3: first $24 \times 3 = 72$. There are $2 + 1 = 3$ decimal places, so $0.24 \times 0.3 = \boxed{0.072}$.

Division with a decimal divisor

Multiply both dividend and divisor by the same power of 10:

$$4.68 \div 0.13 = 468 \div 13 = 36.$$

Why it works: Multiplying the top and bottom of the same fraction by 100 does not change its value.

Does multiplication always make bigger numbers?

- With positive numbers greater than 1: $3 \times 1.2 = 3.6$ (larger).
- With a factor between 0 and 1: $3 \times 0.2 = 0.6$ (smaller).
- Division by a positive number below 1 can make a bigger number: $3 \div 0.2 = 15$.

Terminating and repeating decimals

$\frac{1}{8} = 0.125$ terminates, while $\frac{1}{3} = 0.333\ldots$ repeats.
In $\frac{1}{7} = 0.\overline{142857}$ the block 142857 repeats. A decimal may continue indefinitely; do not insert an invented last digit.

Leap years: a decimal application

The textbook uses approximately 365.2422 days for one Earth revolution. Since a calendar counts whole days, it uses extra days in certain years. The Gregorian rule is: leap year when divisible by 4, *except* centuries are not leap years unless divisible by 400.
For 10,000 calendar years (years 1–10,000): $365 \times 10000 + (2500 - 100 + 25) = \boxed{3,652,425}$ days. The textbook approximation gives $365.2422 \times 10000 = \boxed{3,652,422}$ days, a difference of $\boxed{3}$ days. This is a long-term approximation exercise, not a statement that the year is exactly 365.2422 days.

6. Common mistakes, memory tips and self-test

Beware of these:

- $4.23 \times 3.7 \neq 156.51$. Count $2 + 1 = 3$ decimal places: the correct answer is 15.651.
- For $2.46 \div 0.15$, multiply *both* numbers by 100, not only the divisor.
- In money problems, round the *final* answer to paise only when payment requires it.
- For comparisons of packets of different weights, compare cost *per gram*.

Memory card: Multiply: **count places**. Divide: **shift both**. Compare prices: **same units**. Check: quotient × divisor = dividend.

Quick self-test (original practice, not NCERT)

(1) $0.35 \times 0.6 = \boxed{0.21}$. (2) $7.2 \div 0.06 = \boxed{120}$.
(3) $0.75\,\text{kg} = \boxed{750\,\text{g}}$. (4) $8 \div 0.5 = \boxed{16}$.